

Karan Pathak

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EDUCATION



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

August 2024 - July 2028
PHD - EMBEDDED SYSTEMS LAB, EPFL



TECHNISCHE UNIVERSITEIT DELFT

M.Sc. EMBEDDED SYSTEMS (SUMMA CUM LAUDE)
September 2021-August 2023 | GPA: 8.7/10



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER EXCHANGE | GPA: 5.45/6
SWISS EUROPEAN MOBILITY PROGRAM
September 2022-19 February 2023
M.Sc. THESIS - EMBEDDED SYSTEMS LAB, EPFL
September 2022 - August 2023



NATIONAL INSTITUTE OF TECHNOLOGY HAMIRPUR, INDIA

B.TECH. ELECTRONICS AND COMMUNICATION ENGINEERING
2016 | GPA: 9.52/10
Silver Medal
Merit Scholarship

GRANTS AND SCHOLARSHIPS

- Swiss European Mobility Program (2022-2023) Scholarship by EPFL.
- Justus and Louise van Effen Grant 2022-2023 by TU Delft - awarded to top performing students
- Merit Scholarship- Bachelors of Technology, for a period of 4 years covering full living expenses

EXPERIENCE

DOCTORAL CANDIDATE | ESL, EPFL, SWITZERLAND.

- Beyond GPU memory scaling of LLMs using CXL memory expander cards.
- Creating Simulation infrastructure for next generation Interconnects - CXL; funded by Intel.
- Hiding Latency of CXL Memory with emerging programming models.

MTS/R&D ENGINEER | COMPUTE SYSTEM ARCHITECTURE (CSA), INTER-UNIVERSITY MICROELECTRONICS CENTER (IMEC), LEUVEN, BELGIUM. | SEPTEMBER-23 - JULY-24

- Creating simulation and emulation infrastructure for early performance estimation of novel architectures. (gem5, SST, Ramulator)
- Hardware prototyping for machine learning workloads. (Vivado-AMD FPGAs)
- Prototyping State-of-the art accelerators on FPGA.(Chisel)

AWARDS

- Summa Cum Laude** - Masters in Embedded System with specialization in Computer Architecture.
- Silver Medal**-Bachelors of Technology. Ranked second in Electronics and Communication Engineering Department, Batch of 2012-2016.
- Ranked 989** worldwide in "International Mathematics Olympiad"
- Awarded **Gold Medal** in National Mathematics Olympiad India-Secured 94 percent score
- Awarded **Gold Medal**-Ranked 1st in State in Science Olympiad.

PUBLICATIONS

- K.Pathak**, et al., "CXLRAMSim v1.0: System-Level Exploration of CXL Memory Expander Cards," March 2026, <https://arxiv.org/abs/2603.29483>
- Karan Pathak**, et al. 2025. Towards Accurate RISC-V Full System Simulation via Component-Level Calibration. **ACM Trans. Embed. Comput. Syst.** 24, 4, Article 57 (July 2025), 19 pages. <https://doi.org/10.1145/3737876>
- Debjyoti Bhattacharjee, Anmol, Tommaso Marinelli, **Karan Pathak**, Peter Kourzanov, et al. "Full-stack evaluation of Machine Learning inference workloads for RISC-V systems," <https://arxiv.org/abs/2405.15380>
- Karan Pathak**, et al. "Validating Full-System RISC-V Simulator: A Systematic Approach," RISC-V Summit Europe 2023, Barcelona, Spain, 5-9 June, 2023.